**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Programming: Unseen Problems and Debugging

**Name** _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

## Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does WHERE do in an SQL query?

2. When does a compiler translate source code?

A) When a key is pressed
B) Only after the program ends
C) Before the program runs
D) Only while each line runs

3. "A router's main job is executing spreadsheet formulas." Correct this. Explain why.

4. A computer sends data to a printer on its LAN. "A switch sends every unicast frame to every port." Correct this. How should the data reach the printer?

## Today's objectives

Explain the core ideas of programming: unseen problems and debugging using accurate OCR terminology. Apply a clear method to the worked and guided questions.

## Key vocabulary - Read, cover, recall

**Understand the scenario**

Identify input, processing, output and restrictions before coding.

**Write a plan**

Use simple steps, selections and loops before translating into code.

**Trace unseen code**

Record value changes and follow each condition accurately.

**Debug and improve**

Compare expected and actual behaviour, correct the cause and retest.

**Write one key term and its meaning in your own words:**

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Learn and practise

## New learning - Key ideas

| | |
|---|---|
| **Understand the scenario** | Identify input, processing, output and restrictions before coding. |
| **Write a plan** | Use simple steps, selections and loops before translating into code. |
| **Trace unseen code** | Record value changes and follow each condition accurately. |
| **Debug and improve** | Compare expected and actual behaviour, correct the cause and retest. |

## I Do - One worked example

**An unknown program calculates price = quantity * unitPrice but prints unitPrice. Find the fault.**

1. Identify expected output as total price.
2. Trace sample quantity=3, unitPrice=4; price=12.
3. The print statement uses the wrong variable.
4. Change to print(price), retest output 12.

## We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A program should count 1 to 5 but stops at 4. Explain a fix.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Think and write

## Code First - Predict > Write > Test > Improve

**YOUR TASK: Correct the operator in a storage cost-per-GB calculation.**

```
price = 48
capacityGB = 12
costPerGB = price * capacityGB
print(costPerGB)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

## You Do - Independent exam questions

Q2. Explain why a trace is useful for unfamiliar code. [3 marks]

Q3. Correct an algorithm that checks 1<=x<=10 using OR between two comparisons. [2 marks]

Q4. State two steps to take after correcting a logic error. [2 marks]

## Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A trace can help locate the first incorrect variable update. ☐ T ☐ F
2. Identify input, processing, output and restrictions before coding. ☐ T ☐ F
3. Use simple steps, selections and loops before translating into code. ☐ T ☐ F

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

## Do Now - Answers

**Q1** It selects rows that meet a condition. ☐

**Q2** C - Before the program runs ☐

**Q3** It forwards traffic between different networks using IP addressing. ☐

**Q4** A switch uses MAC addresses to send data to the right connected device. It does not normally send each known unicast frame to every port. ☐

## Code First - Model

```
price = 48
capacityGB = 12
costPerGB = price / capacityGB
print(costPerGB)
```

**Output: 4**

## We Do - Check the model

1. Test both first and last values. ☐

## Exit Ticket - Check

**1. TRUE 2. TRUE 3. TRUE**

## You Do - Self-mark

**Q2** It makes executed steps and changing variable values explicit, exposing incorrect assumptions or logic. ☐

**Q3** Use x>=1 AND x<=10. ☐

**Q4** Run/test with expected normal and boundary cases, and compare outcomes. ☐

## Try the question again

**1. Copy the FULL question you found most difficult:**

**2. In GREEN PEN, write a complete new answer:**